

DASE 7506 Mini Project 1 Report

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Objective

Improve the baseline GPT model provided to achieve the lowest testing result possible, by changing the model structure and/or training procedure.

Method

The experiment is done by running the provided commands after tweaking configs/modified.json and student.py.

Addition of dropout

After gaining an understanding of the model's structure, it is noticed that the dropout section from a usual transformer block is not present. As a result, dropout is added and compared for any possible improvements.

More specifically, the dropout is added after both attention and MLP, with dropout rate of 0.1.

Swapping the activation function

As an initial attempt, the activation function for MLP is changed from GELU to ReLU.

Increase in training steps

Since this would not affect the speed during testing, the training step is increased to both 2400 and 4800 for comparison.

Other attempts

The number of heads and depths were also changed but seemed to produce very limited effect throughout the experiments.

In an attempt to decrease training time, the upscaling in MLP was halved to 2. Obviously, the results went down, although not as drastically as expected.

Results & Analysis

Training Steps	Baseline	ReLU	Dropout
1200	2.101	2.052	2.135
2400	1.909	1.873	1.940
4800	1.777	1.767	1.845

More training yields better results

As seen in the table, increasing the amount of training steps contributes to most of the decrease in BPB score. As the loss start fluctuating around 3.8 beyond 3000 steps, it is likely that further increasing training steps would yield less prominent effect.

Dropout causes negative effect

Under all conditions, adding dropout only increases BPB result, indicating downgrade to the model. Since the model has limited width and depth, introducing dropout seems to lose important attention information, instead of limiting overfitting.

ReLU achieves minor improvements

Surprisingly, replacing GELU with ReLU in fact slightly improves the behaviour, possibly also due to the small scale and training count of the model.

It is also noticeable that the difference decreases as training steps increases. At higher training steps, it is possible that GELU would outperform ReLU eventually.

As ReLU is less complicated than GELU, a minor increase in model speed is likely present as well, although only a 0.25 second difference is present for testing on CPU.

AI Usage Declaration

AI was relied on heavily during the initial period of trying to understand the project's structure. Copilot in VSCode (which I believe was GPT Luna) was used.

Later on, discussions were made between GPT 5.4 mini to find explanations for some of the results.